

UET University Peshawar

Abbottabad Campus



Lab Report No: 10

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Class No: 21

Course: SE-305L Computer Communications and Networks

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Lab 10: Analyzing Local Area Network (LAN) Traffic Patterns in Wireshark

Objective:

In this lab, students will analyze communication patterns within a Local Area Network (LAN) using Wireshark. By the end of this lab, students will:

- Identify different types of LAN traffic (broadcast, multicast, unicast).
- Examine ARP, ICMP, and local TCP/UDP communications.
- Understand MAC and IP addressing in LAN environments.
- Apply filters to isolate and interpret specific LAN interactions.
- Measure traffic patterns using statistics tools.

Lab Tasks Overview:

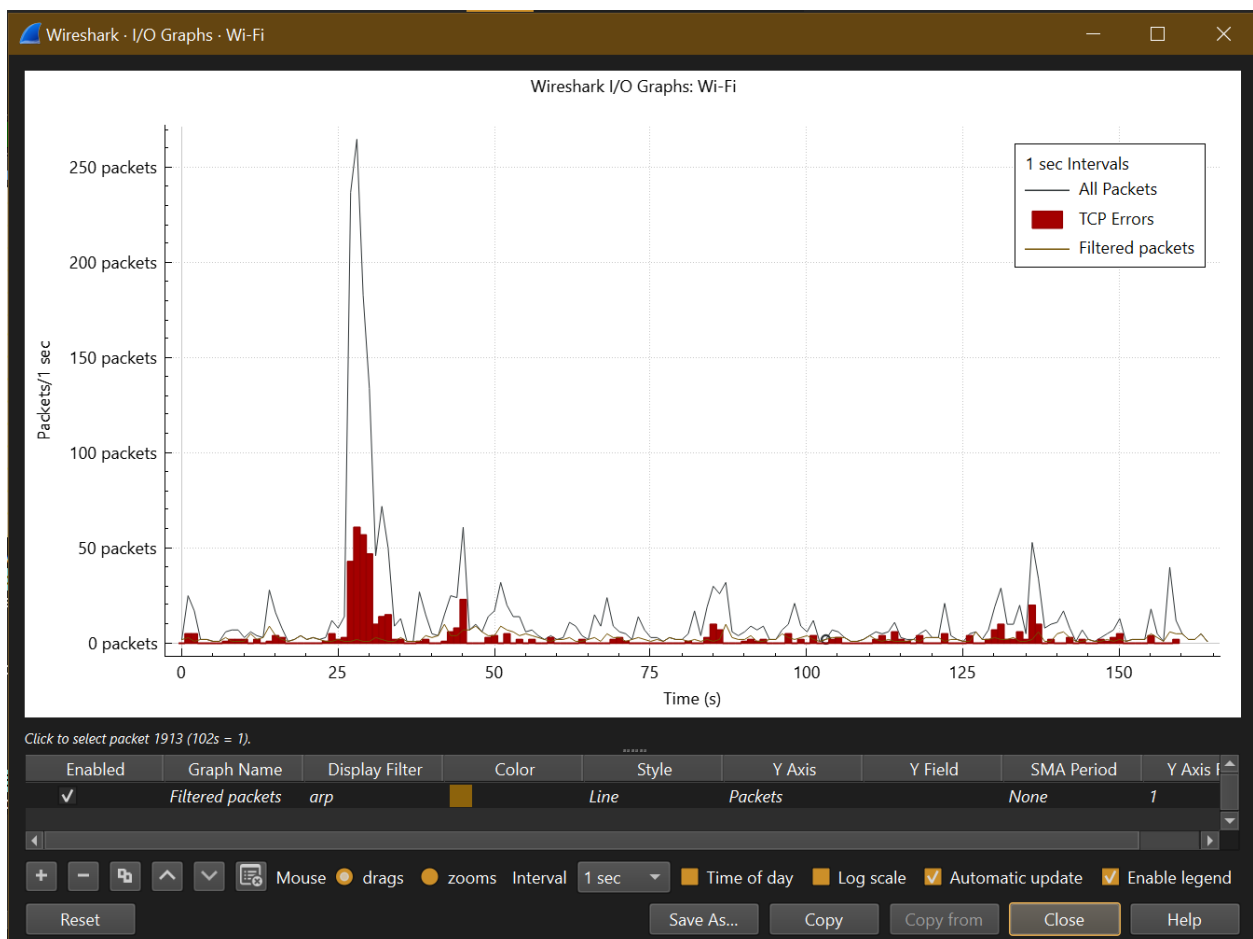
1. Capture LAN Traffic
2. Analyze ARP and Broadcast Traffic
3. Observe ICMP and Device Discovery
4. Analyze MAC vs IP Addressing
5. Generate and Interpret LAN-Specific IO Graphs 6. Export Summary and Report

Task 1: Capture LAN Traffic Instructions:

1. Launch Wireshark.
2. Select the active interface (Wi-Fi/Ethernet).
3. Start the packet capture.
4. While capturing:
 - o Open Command Prompt/Terminal.
 - o Ping the default gateway (type ipconfig in the command prompt, to look for default gateway).
 - o Ping another device (computer) on the same LAN.
5. Stop the capture after 30–60 seconds.

No.	Time	Source	Destination	Protocol	Length	Info
1779	87.252375	Apple_c6:cb:cc	Broadcast	ARP	60	ARP Announcement for 192.168.30.147
1789	87.416515	Apple_c6:cb:cc	Intel_5f:e7:bd	ARP	60	192.168.30.147 is at 84:ad:8d:c6:cb:cc
1797	87.564752	Apple_c6:cb:cc	Broadcast	ARP	60	Who has 192.168.30.2? Tell 192.168.30.147
1804	88.477637	Apple_c6:cb:cc	Broadcast	ARP	60	Who has 192.168.30.2? Tell 192.168.30.147
1806	88.783327	Apple_c6:cb:cc	Broadcast	ARP	60	ARP Announcement for 192.168.30.147
1809	89.090539	Apple_c6:cb:cc	Broadcast	ARP	60	Who has 192.168.30.2? Tell 192.168.30.147
138	16.896287	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.185? Tell 192.168.30.2
144	19.083131	Cisco_07:83:47	Intel_5f:e7:bd	ARP	60	52.55.101.50 is at cc:ef:48:07:83:47
1425	50.692075	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.68? Tell 192.168.30.2
1477	52.542922	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.185? Tell 192.168.30.2
1480	52.842503	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.68? Tell 192.168.30.2
1513	55.166560	Cisco_07:83:47	Intel_5f:e7:bd	ARP	60	52.55.101.50 is at cc:ef:48:07:83:47
1526	57.449286	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.68? Tell 192.168.30.2
1566	66.051171	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.68? Tell 192.168.30.2
1699	84.176622	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.68? Tell 192.168.30.2
1801	87.864517	Cisco_07:83:47	Broadcast	ARP	60	Who has 192.168.30.185? Tell 192.168.30.2

Frame 110: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Dev
 Ethernet II, Src: f6:bc:3f:68:b2:d8 (f6:bc:3f:68:b2:d8), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 Source: f6:bc:3f:68:b2:d8 (f6:bc:3f:68:b2:d8)
 Type: ARP (0x0806)
 Trailer: edc31db4c01001e5172300000101080aacf
 Address Resolution Protocol (request)



Checkpoint Questions:

- Q1. What types of traffic do you observe when you ping the default gateway?

Answer: When pinging the default gateway, the primary type of traffic observed is **ARP (Address Resolution Protocol)**. The image clearly shows a Wireshark capture filtered for "arp" packets, and the main window displays numerous ARP requests and announcements. While ping typically involves ICMP (Internet Control Message Protocol) traffic, the provided image focuses on the ARP aspect of the communication that occurs *before* or *during* the ping process to resolve IP addresses to MAC addresses.

- Q2. Which IP and MAC addresses are involved in the captured communication (check ARP packets)?

Answer: By examining the ARP packets, we can identify the following IP and MAC addresses involved:

- **Sender/Source IP Addresses:**

- 192.168.30.147 (This IP address is frequently the target of "ARP Announcement" and "ARP Who has" requests).
- 192.168.30.22
- 192.168.30.27
- 192.168.30.18
- 192.168.30.50
- 192.168.30.2
- 192.168.30.195

- **Sender/Source MAC Addresses:**

- Apple_c6:cb:cc (e.g., in packet 1789, 1797, 1809, 1886)
- Cisco_87:83:47 (e.g., in packet 139, 1425, 1477, 1480)
- Intel_5f:e7:bd (observed as a destination MAC, but also implicitly involved in replies)
- b4:ad:80:c6:cb:cc (specifically mentioned as the MAC address for 192.168.30.147 in an ARP announcement detail)
- cc:ef:48:87:83:47 (specifically mentioned as the MAC address for 192.168.30.50)

- **Target IP Addresses:**

- 192.168.30.147 (frequently the target of "Who has" requests)
- 192.168.30.22
- 192.168.30.18
- 192.168.30.2
- **Target MAC Addresses:**
 - 00:00:00:00:00:00 (for ARP requests, indicating the MAC address is unknown)
 - Intel_5f:e7:bd (frequently a destination MAC, suggesting it's the MAC of the default gateway or another device on the network)
 - Broadcast (for ARP requests)

Task 2: Analyze ARP and Broadcast Traffic Instructions:

1. In the filter bar, type arp and press Enter.
2. Locate an ARP Request and corresponding ARP Reply.
3. Click and examine:
 - Source/Destination MAC addresses.
 - IP-to-MAC resolution.

The screenshot displays the Wireshark interface with the filter 'arp' applied. The packet list shows several ARP requests and announcements. The selected packet (No. 1779) is an ARP request from Apple_c6:cb:cc to Intel_5f:e7:bd. The packet details pane shows the 'Address Resolution Protocol (ARP)' section with fields like Hardware type, Protocol type, Opcode, and MAC/IP addresses. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1779	87.252375	Apple_c6:cb:cc	Broadcast	ARP	60	ARP Announcement for 192.168.30.147
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Packet 1779 details:

```

Type: ARP (0x0806)
[Stream index: 47]
Padding: 00000000000000000000000000000000
Address Resolution Protocol (ARP Announcement)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  [Is gratuitous: True]
  [Is announcement: True]
  Sender MAC address: Apple_c6:cb:cc (84:ad:8d:c6:cb:cc)
  Sender IP address: 192.168.30.147
  Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)
  Target IP address: 192.168.30.147
  
```

Packet bytes:

```

0000 ff ff ff ff ff ff 84 ad 8d c6 cb cc 08 06 00 01
0010 84 ad 06 04 00 01 84 ad 8d c6 cb cc c9 a8 1e 93
0020 00 00 00 00 00 00 c0 a8 1e 93 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00
  
```

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
2226	138.528220	172.217.17.78	192.168.30.166	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=113 (request in 2225)
2236	139.539922	172.217.17.78	192.168.30.166	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=113 (request in 2235)
2249	140.556813	172.217.17.78	192.168.30.166	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=113 (request in 2246)
2260	141.575308	172.217.17.78	192.168.30.166	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=113 (request in 2259)
1796	87.443580	192.168.30.147	192.168.30.166	ICMP	70	Destination unreachable (Port unreachable)
1648	73.740018	192.168.30.154	192.168.30.166	ICMP	70	Destination unreachable (Port unreachable)
1688	82.053251	192.168.30.155	192.168.30.166	ICMP	120	Destination unreachable (Port unreachable)
1689	82.092835	192.168.30.155	192.168.30.166	ICMP	120	Destination unreachable (Port unreachable)
2224	138.485445	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (no response found!)
2225	138.485452	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 2226)
2234	139.497116	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (no response found!)
2235	139.497132	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 2236)
2245	140.514238	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (no response found!)
2246	140.514248	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 2249)
2258	141.532387	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (no response found!)
2259	141.532392	192.168.30.166	172.217.17.78	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 2260)

Frame 1689: 120 bytes on wire (960 bits), 120 bytes captured (960 bits) on interface...

Ethernet II, Src: f2:55:3d:55:c7:24 (f2:55:3d:55:c7:24), Dst: Intel_5f:e7:bd (04:e8:b9:5f:e7:bd)

Destination: Intel_5f:e7:bd (04:e8:b9:5f:e7:bd)

Source: f2:55:3d:55:c7:24 (f2:55:3d:55:c7:24)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 192.168.30.155, Dst: 192.168.30.166

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

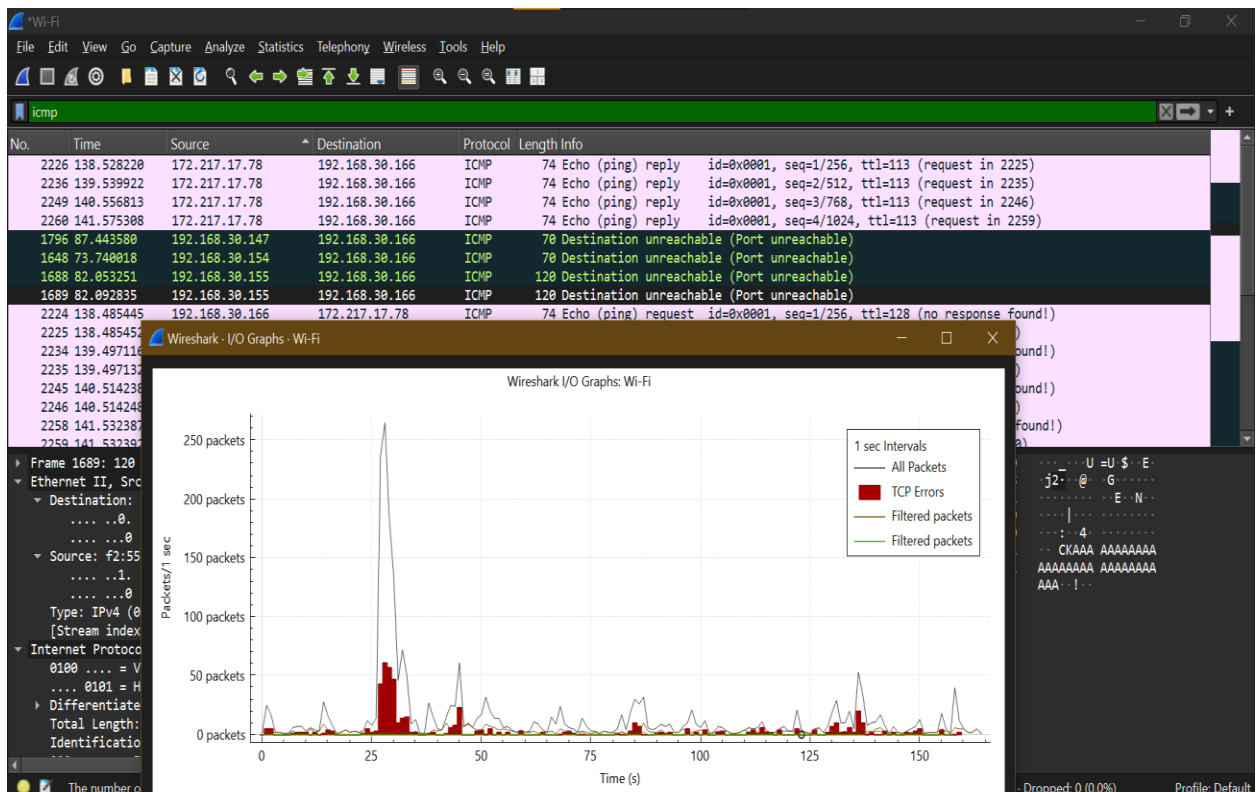
Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 106

Identification: 0x32fa (13050)

Internet Control Message Protocol Protocol

Packets: 2411 - Displayed: 24 (1.0%) - Dropped: 0 (0.0%) Profile: Default



- Q3. What is the purpose of an ARP Request?

Answer: ARP packet dissection, the purpose of an ARP Request is to **resolve an IP address to its corresponding MAC (hardware) address on a local network segment.**

Specifically, the "ARP Request" in the image states "Who has 192.168.30.147? Tell 192.168.30.147." This clearly indicates that the sender (with IP 192.168.30.147) is trying to find out the MAC address of a device with the IP address 192.168.30.147 (it seems to be an ARP announcement or a gratuitous ARP, as the sender and target IP are the same, which is used to announce one's own MAC address or detect duplicate IPs). In a typical ARP request for another host, the "Target IP address" would be the IP of the device whose MAC address is being sought, and the "Target MAC address" would be all zeros (00:00:00:00:00:00).

- Q4. How does the ARP Reply differ from the Request?

Answer: While the provided images mostly show ARP Announcements/Requests, a typical ARP Reply differs from an ARP Request in the following key ways:

- **Opcode:** An ARP Request has an opcode of 1 (Request), while an ARP Reply has an opcode of 2 (Reply).
 - **Destination MAC Address:** An ARP Request is typically sent as a **broadcast** to the MAC address FF:FF:FF:FF:FF:FF (as seen in the "lab10.png" image for several packets). This means all devices on the local network segment receive the request. An ARP Reply, however, is a **unicast** packet, meaning it is sent directly to the MAC address of the device that sent the original ARP Request.
 - **Target MAC Address in Packet Content:** In an ARP Request, the "Target MAC address" field is typically all zeros (00:00:00:00:00:00) because the sender doesn't know it. In an ARP Reply, the "Target MAC address" field contains the MAC address of the device that *sent* the original request, and the "Sender MAC address" field contains the MAC address of the device *responding* to the request (the one whose MAC address was asked for).
- Q5. Why is ARP considered a broadcast protocol?
 - Answer: Query successful

Let's analyze the provided Wireshark captures to answer your questions about ARP.

Q3. What is the purpose of an ARP Request?

Based on the "lab10.png" image, which shows an ARP packet dissection, the purpose of an ARP Request is to **resolve an IP address to its corresponding MAC (hardware) address on a local network segment.**

Specifically, the "ARP Request" in the image states "Who has 192.168.30.147? Tell 192.168.30.147." This clearly indicates that the sender (with IP 192.168.30.147) is trying to find out the MAC address of a device with the IP address 192.168.30.147 (it seems to be an ARP announcement or a gratuitous ARP, as the sender and target IP are the same, which is used to announce one's own MAC address or detect duplicate IPs). In a typical ARP request for another host, the "Target IP address" would be the IP of the device whose MAC address is being sought, and the "Target MAC address" would be all zeros (00:00:00:00:00:00).

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Q5. Why is ARP considered a broadcast protocol?

ARP is considered a broadcast protocol primarily because of the **ARP Request phase**.

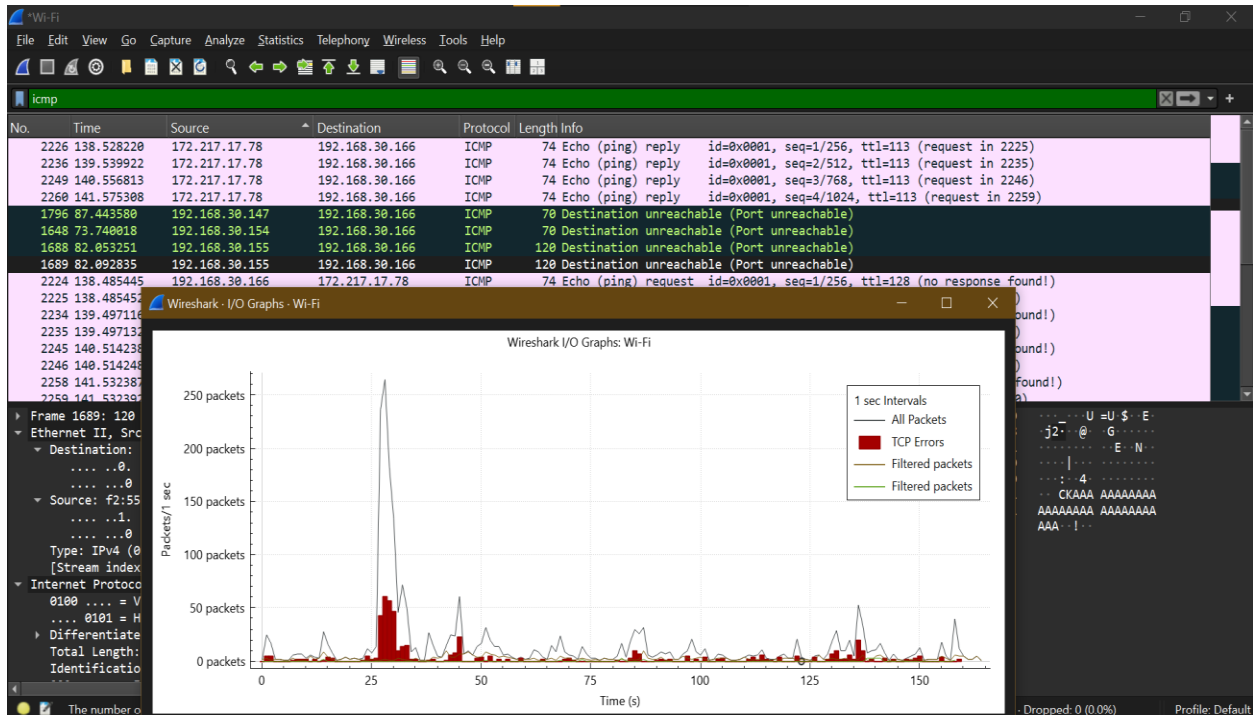
As seen in the "lab10.png" capture, many ARP packets have a "Destination" listed as "Broadcast". When a device needs to find the MAC address for a specific IP address and it doesn't have it in its ARP cache, it sends an ARP Request packet with a **broadcast MAC address (FF:FF:FF:FF:FF:FF)** in the Ethernet header. This ensures that every device on the local network segment receives and processes the ARP Request. The device that owns the requested IP address will then send a unicast ARP Reply directly back to the requesting device.

Task 3: Observe ICMP and Device Discovery Instructions:

1. Apply the filter icmp.
2. Identify ICMP Echo Requests and Replies (e.g., ping responses).

3. Determine the round-trip time (RTT) from the packet details.

4. Repeat by pinging another device on the same LAN and observe the new packets.



```
C:\Users\PMLS>config facebook.com
'config' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\PMLS>ping facebook.com

Pinging facebook.com [157.240.227.35] with 32 bytes of data:
Reply from 157.240.227.35: bytes=32 time=41ms TTL=49
Reply from 157.240.227.35: bytes=32 time=41ms TTL=49
Reply from 157.240.227.35: bytes=32 time=42ms TTL=49
Reply from 157.240.227.35: bytes=32 time=42ms TTL=49

Ping statistics for 157.240.227.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 41ms, Maximum = 42ms, Average = 41ms
```

Checkpoint Questions:

- Q6. What is the purpose of ICMP in LANs?
 - Answer: **Echo (ping) requests and replies:** These are the most common uses of ICMP for basic connectivity testing.

- **Destination unreachable (Port unreachable):** This indicates that a packet reached the destination host, but the application or service it was intended for was not available on that port. This is an error message.

In summary, ICMP helps devices communicate network-layer information, including:

- **Confirming reachability:** As seen with ping requests/replies.
 - **Reporting errors:** Such as "Destination unreachable" when a host or service cannot be reached.
 - **Controlling flow and congestion:** Though less common in basic ping scenarios, ICMP also has messages for source quench and redirect.
- Q7. How can ICMP be used to test device availability?

ICMP is extensively used to test device availability primarily through the "**ping**" utility, which generates **ICMP Echo Request and Echo Reply messages**.

Here's how it works, as illustrated by the "lab10.22.png" and "lab10.2.png" captures:

1. **Sending an Echo Request:** Your device (e.g., 172.217.17.78 or 192.168.30.155) sends an **ICMP Echo Request** packet to a target device (e.g., 192.168.30.166 or 192.168.30.147). These packets are visible in the Wireshark capture as "Echo (ping) request."
2. **Receiving an Echo Reply:** If the target device is operational and reachable, it responds with an **ICMP Echo Reply** packet back to the sender. These are visible as "Echo (ping) reply" in the capture.

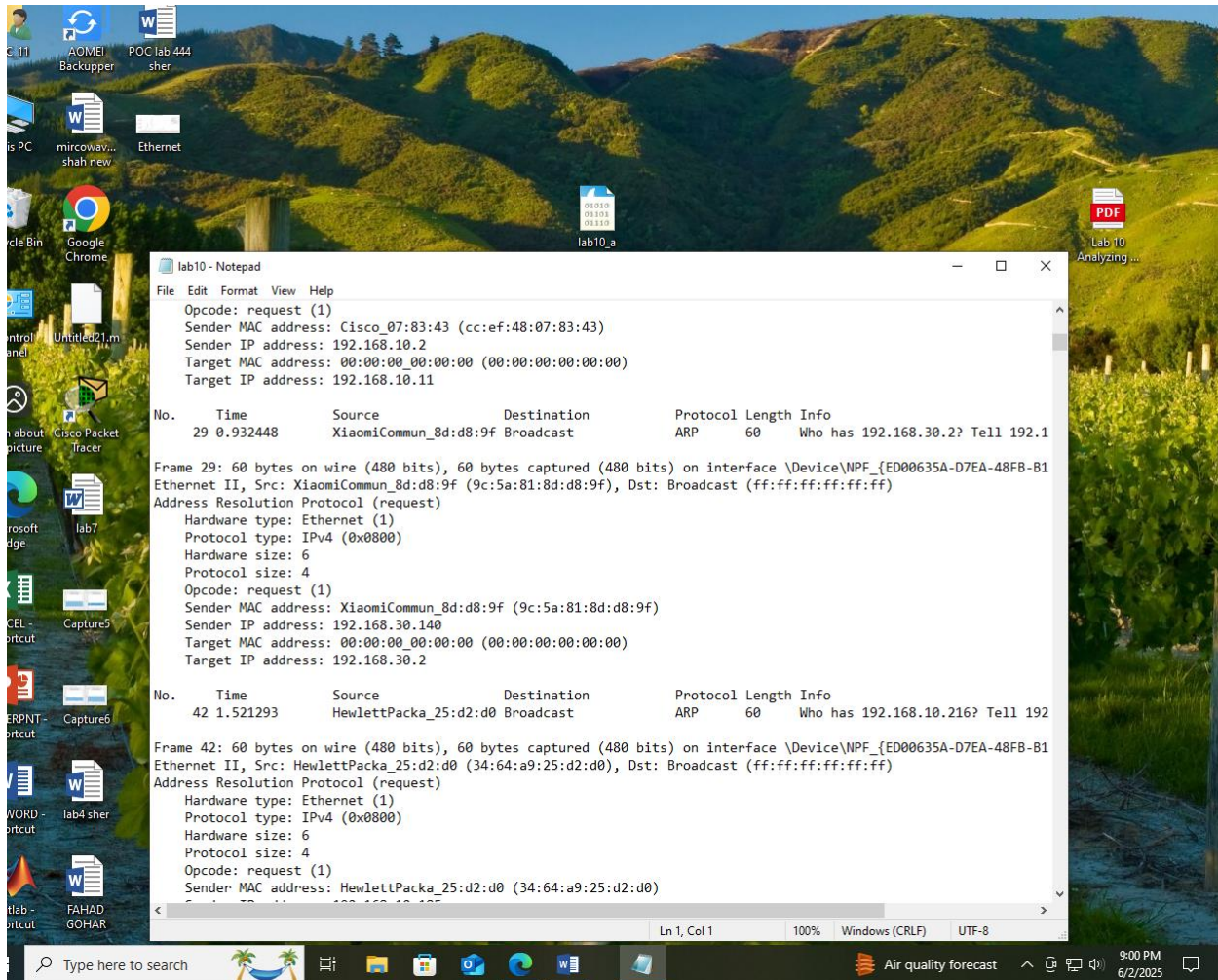
By observing whether an Echo Reply is received within a reasonable timeframe after sending an Echo Request, you can determine:

- **If the target device is online and functioning at the network layer.**
- **If there is a working network path between your device and the target device.**
- **The round-trip time (RTT),** which is the time it takes for the request to go to the target and the reply to come back. This indicates network latency.

Task 4: Analyze MAC vs IP Addressing Instructions:

1. Select an ARP or ICMP packet.
2. Compare the Ethernet (Layer 2) and IP (Layer 3) addresses.
3. Note the use of MAC addresses within the LAN and IP addresses for routing.

4. Write down your system's MAC and IP address.



Checkpoint Questions:

- Q8. How do MAC and IP addresses work together in a LAN?

Answer: MAC (Media Access Control) addresses and IP (Internet Protocol) addresses work together in a layered fashion to enable communication within a LAN and beyond:

- **MAC Addresses (Layer 2 - Data Link Layer):**
 - **Purpose:** MAC addresses are physical hardware addresses uniquely assigned to network interfaces (like Ethernet cards, Wi-Fi adapters) by the manufacturer. They are used for **local communication within a single network segment (LAN)**.
 - **Functionality:** When devices communicate on a LAN, they use MAC addresses to identify each other directly. Data frames are addressed to the destination MAC address.

- **Example from captures:** In lab10.png and image_05ad92.jpg, you see MAC addresses like Apple_C6:cb:cc, Intel_5f:e7:bd, Cisco_87:83:43, XiaomiCommun_8d:d8:9f, and HewlettPacka_25:d2:d0 as Source and Destination addresses in ARP packets.
- **IP Addresses (Layer 3 - Network Layer):**
 - **Purpose:** IP addresses are logical addresses assigned to devices by a network administrator or DHCP server. They are used for **identifying devices across different networks (including the internet) and for routing traffic.**
 - **Functionality:** IP addresses define the source and destination of data packets in a network. When data needs to be sent to a device on a *different* network, routers use IP addresses to forward the packet.
 - **Example from captures:** In lab10.png and image_05ad92.jpg, you see IP addresses like 192.168.30.147, 192.168.30.2, 192.168.10.11, and 192.168.10.216 being queried or announced.

How they work together:

1. **ARP (Address Resolution Protocol) is the bridge:** When a device wants to send data to another device on the *same local network*, it knows the destination's IP address (Layer 3). However, to actually put the data on the wire, it needs the destination's MAC address (Layer 2). This is where ARP comes in.
 - As seen in lab10.png and image_05ad92.jpg, an **ARP Request** is broadcast to ask, "Who has this IP address? Tell me your MAC address."
 - The device with that IP address sends an **ARP Reply** containing its MAC address.
2. **Encapsulation:** Once the sending device knows both the IP and MAC addresses of the destination (or the default gateway if the destination is on another network), it encapsulates the IP packet within an Ethernet frame.
 - The **IP header** contains the Source IP and Destination IP addresses.
 - The **Ethernet header** contains the Source MAC and Destination MAC addresses.
3. **Local Delivery:** The Ethernet frame, now with the correct destination MAC address, is sent over the local network segment. Switches use MAC addresses to forward the frame only to the port connected to the destination device.
4. **Routing (for external networks):** If the destination IP address is on a *different* network, the sending device sends the packet to its **default gateway's MAC address**. The router

(default gateway) then uses the destination IP address to route the packet to the next hop until it reaches the destination network. On each network segment, ARP is used to resolve the next hop's MAC address.

- Q9. Can two devices have the same IP address in a LAN? What happens if they do?

No, two devices should not have the same IP address in a LAN. IP addresses are designed to be unique identifiers for devices on a given network segment.

What happens if they do (IP address conflict):

If two devices are configured with the same static IP address, or if a DHCP server mistakenly assigns the same IP address to two different devices, an **IP address conflict** occurs. This leads to severe network communication problems, including:

1. **Intermittent Connectivity Issues:** Both devices will experience unreliable network access.
 - Packets intended for that IP address might be delivered to one device at one moment and the other device at another moment, leading to dropped connections, timeouts, and applications failing.
 - As seen in lab10.png, there are "ARP Announcement for 192.168.30.147" packets. Gratuitous ARP (where a device announces its own IP and MAC) can sometimes be used to detect duplicates, as a device will send out an ARP request for its *own* IP and if it receives a reply, it knows there's a conflict.
2. **ARP Cache Poisoning/Instability:**
 - Devices on the network will receive conflicting ARP replies (different MAC addresses for the same IP address). Their ARP caches will constantly be updated with incorrect or rapidly changing information, leading to confusion about which MAC address belongs to the shared IP.
 - This is evident in lab10.png where you see multiple ARP announcements or requests for the same IP (192.168.30.147 or 192.168.30.2). While some might be legitimate (e.g., announcing a new MAC after an adapter change), a rapid succession of different MAC addresses for the same IP is a strong indicator of a conflict.